

Strengthening Community Data: Towards Pervasive Participation

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ABSTRACT

Data, big data, and data analytics are contemporary touchstones for rational and technocratic thinking about human activity. Data also provide a new foundation and focus for community innovation, and new concrete possibilities for enhancing community activities, experiences, and civic participation. We present and discuss a series of community design scenarios envisioning how community data can be produced, gathered, analyzed, interpreted and used by community members to enhance community awareness and engagement, to provide new opportunities for learning and development among community members, and to increase the transparency, effectiveness and fairness of local government. Our argument is that community data can be seen as a vehicle not only for more pervasive participation in local communities but also for richer, better-grounded, and more effective participation.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative content creation; Social engineering (social sciences); Ubiquitous computing**; • **Information systems** → *Federated databases*; • **Social and professional topics** → *Governmental regulations*;

KEYWORDS

Community data, data literacy, pervasive participation

ACM Reference Format:

John M. Carroll, Jordan Beck, Shipi Dhanorkar, Jomara Binda, Srishti Gupta, and Haining Zhu. 2018. Strengthening Community Data: Towards Pervasive Participation. In *dg.o '18: dg.o 2018: Proceedings of the 19th Annual International Conference on Digital Government Research, May 30-June 1, 2018, Delft, Netherlands*, Anneke Zuiderwijk and Charles C. Hinnant (Eds.). ACM, New York, NY, USA, Article 4, 9 pages. <https://doi.org/10.1145/3209281.3209312>

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dg.o'18, June 2018, Delft, NL

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ACM ISBN 978-1-4503-6526-0/18/05...\$15.00
<https://doi.org/10.1145/3209281.3209312>

1 COMMUNITY INNOVATION: IDENTITY WORK, PERVASIVE PARTICIPATION, AND SOCIAL SUPPORT

Local community is an abiding and pervasive social structure. It is the foundation for resilient societies. Moreover, throughout modern history, community has served as a crucible for integrating socio-technical change [31, 36].

Today, communities systematically drive innovation in information infrastructures to enable policy and planning objectives [11, 14, 24]. Innovation strengthens the traditional foundations of community by providing new ways to develop and express personal and collective identity, new ways to be aware of and participate in local activities, and new ways to share social support [5, 23].

In this paper, we examine needs and opportunities for citizens to better understand and contribute to data-driven civic participation as a focus for community innovation. On the one hand, today's citizens need to achieve some level of data literacy. They need to have facility in understanding and using data, and they need to be able to think critically and creatively about the many uses of data they might encounter.

A framing assumption in this paper is that data literacy is a worthwhile goal. We argue that community members ought to possess a basic level of competency in data collection and analysis. This puts us into opposition with others who may believe that citizens should not have to learn about particular kinds of data or to develop certain competencies in working with data in order to use and benefit from it.

In our view, data literacy will help citizens identify relevant data, analyze and interpret data, participate more actively as community members, and use data in everyday civic contexts. The use of data and data-driven argumentation to shape decisions and policies at the local level is an opportunity to strengthen local democracy and democratic practices in general. These threads converge in the transformative possibility that data-enabled citizens could more constructively and effectively participate in and shape local governance that is itself data-oriented.

Local governance has often failed through conflicts grounded in irreconcilable judgment and self-interest [9]. While data may not be a panacea for human conflict, they create an opportunity for more rational discourse to take place. At the very least, data are a shared community resource that has been under leveraged.

We refer to data pertaining to a community and its locale, that is, data gathered, analyzed, interpreted, and used by members of a local community, as *community data*. Our interpretation of data is broad and inclusive; we consider data to include environmental measures (e.g. water and air quality) as well as photographs and short narratives about personal experiences. Through community data, the community describes the community to itself in order to understand its past, regulate its present, and deliberate its future. Community data is thus *hyperlocal* [6].

There has always been community data, but the extent to which community members could participate directly in data-driven public processes has been limited. Even now, data gathering by municipal governments and community non-profit groups may treat citizens as objects of study rather than as partners or co-investigators. And while there are efforts to facilitate citizen engagement with data around public policy and decision-making [13, 27], there is more work to do. One issue in the open data literature, for example, has to do with getting different groups of people to engage with and use data that is open to them.

In a democratic society, such social dynamics form an important factor and lead to community engagement and participation. The idea of community engagement and participation in various community-related matters has long been advocated by researchers. One obvious rationale behind this is that it strengthens the relationship between citizens and their local government, which in-turn leads to a stronger democratic society. However, in some scenarios, such community engagements often turn out to be arduous because of conflicts within the community or disillusionment created by the domineering ‘elitist’ of the community, who take the role of decision-makers [17].

Nevertheless, the advancement of information and communication technologies has enabled a new data-driven society which has helped reduced this gap between the community’s elitist bureaucrats and the rest of the citizens. With the advent of new computational methods and tools such as technologies to manage and analyze Big Data, the process of evidence-based decision making has become much more efficient. Hence, our municipalities have also started using these data-driven methods for decision-making [19, 22].

Current information infrastructures, digital devices and sensors can empower citizens to initiate data-driven investigations of community concerns, such as: heritage protection, water quality management, energy regulation, and public by-way safety. Citizens can identify pertinent issues and research questions, coordinate with other citizens, gather and publish data sets online, and moderate community discussions and deliberations.

In the balance of this paper we present envisionments of pervasively participatory community data activities. Our objective is to better understand what it would take to implement such design proposals at scale, what it would mean to participate in such activities to various community stakeholders, who would benefit and how.

We organize our presentation around three main themes:

- (1) community values and heritage,
- (2) community learning and development,
- (3) community deliberation, decision-making, and governance.

We end with a discussion of what community data can do for its stakeholders and possible steps to take to strengthen its utility and value.

2 COMMUNITY VALUES AND HERITAGE

Community is a key social institution for developing and expressing personal and collective identity, being aware of and participating in local activities, and sharing social support [23]. Hester [18] vividly theorized these on-going projects of a community as a “sacred structure” of specialized activity hubs, places, overlaying a mundane physical space. An important kind of community data, with respect to values and heritage, is the meanings constructed and evoked by community places.

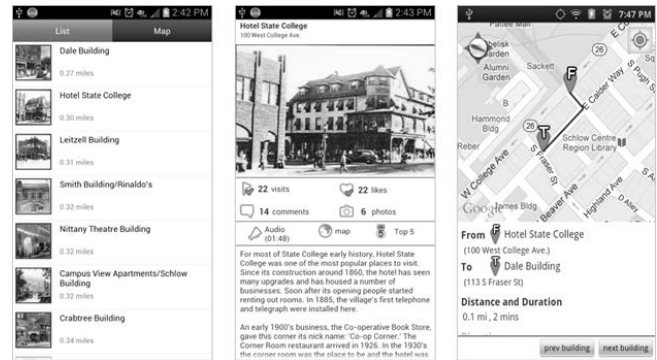


Figure 1: Screenshots from Lost State College mobile app (Han et al., [15])

Lost State College ([15]) was a community heritage app inspired by Hester’s concept of sacred structure (Figure 1). The app contained antique photographs of historically significant locations around the city center of State College, Pennsylvania, USA. The person using the app could view photos of historically significant sites, read about the places depicted, and engage in various social media interactions when at those places. Thus, the person would be present in and see the current place in the world and simultaneously experience its history through the app’s information and the contributions of other users. This example reinforces three crucial points about community data:

- (1) Data can be made to codify the community to enhance visibility and discussability among its members. As we were developing and investigating Lost State College, we were impressed at how ordinary the sacred structure of a community could be. Community members walk by places with historical significance every day, and they rarely take notice.
- (2) Neighbors and fellow citizens contribute community data. This means that the data-makers are a mixture of familiar people and strangers, although they are all members of the same community. The meaning of the data is thus made from different perspectives; only some of which may be familiar. Making sense of the data entails understanding these different perspectives.
- (3) Personal experiences, meanings, and contributions are crucial parts of the data. Lost State College promotes public

sharing of personal experiences tied to community places that other citizens have experienced in unique ways. Thus, just opening up the app creates the potential for novel, interesting ways of contributing to community.

The mobile app distinguishes public space from private space, and, thus, while community members may share private experiences and perspectives, they do so in a public forum about public spaces. The act of contributing to Lost State College is, arguably, inherently communal. In addition, Lost State College evokes interesting questions about the distinction between private and public roles. When is it good to assume a public role? What distinguishes public roles from private ones? How does enacting a private role in a communal setting undermine a sense of community?

Mobile apps have crucial affordances that support activities designed to probe, codify, and share the sacred structure of a community. For example, because of its mobility, Lost State College users experienced different locations in real space and time as they investigated the meanings of these places and reflected on their role as community members. However, mobile technology is not a requirement for creating hyperlocal community data about values and heritage. It is also possible to achieve similar outcomes with web applications.

Blacksburg Nostalgia [8] was an early, web-based system that provided an open forum for residents of Blacksburg, Virginia, USA, to post photos and share stories about what Blacksburg was like in the 1960s (Figure 2). The project was a long-term, participatory design investigation carried out with local elders interested in exchanging stories about local life during their young adulthood.

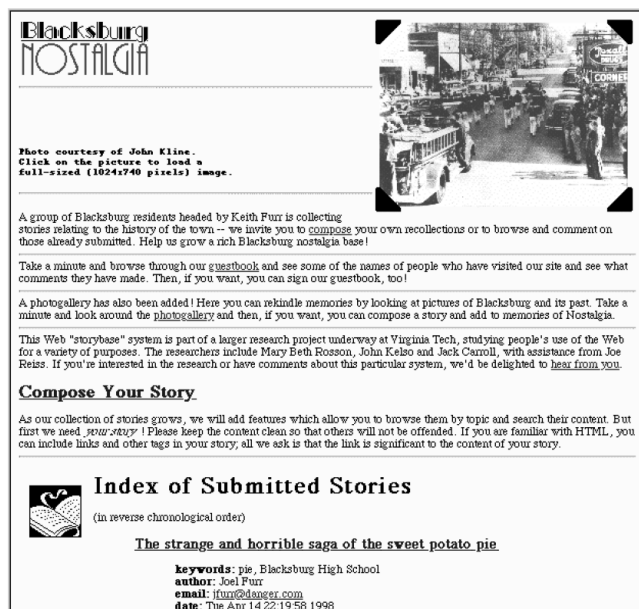


Figure 2: Blacksburg Nostalgia homepage (Carroll et al. [8])

People contributed different kinds of stories to Blacksburg Nostalgia, such as: where train stations and tracks used to be, and when significant weather events (e.g., large snowfalls) took place. Moreover, the system enabled people comment on these different stories;

sharing unique perspectives and memories of how certain events played out or the impact these events may have had on the town.

We see these comments as collective meaning making, reinforcing the notion that community is a tapestry of perspectives and life experiences. Stories on Blacksburg Nostalgia evoked participation from older residents, younger residents, as well as past residents who had since moved out of the community.

Some of the places described in the stories were historical in a conventional sense, such as the location and later transformation of the Huckleberry Train. Other places seemed to be a part of the sacred structure only because of what had happened to a resident there. For example, one resident shared a story about the ability to sit on top of a particular light pole because an unusually heavy snowfall. The light pole itself was otherwise historically insignificant to the community – it was not the first light pole in Blacksburg, it did not serve as the site for an act of civil resistance, and so forth.

An important distinction between Blacksburg Nostalgia and Lost State College is that Blacksburg community members were empowered to bestow historical significance to particular places such as the light pole. In Lost State College, the sacred places were selected in consultation with the State College Historic Resources Commission, which is a top-down, normative view of heritage as places and events anointed by history. We do not mean to suggest that one way of establishing historical significance is better than the other. Instead, we make this distinction to draw attention to the way different notions of heritage inform different practices of gathering and making sense of community data.

In an ongoing project called Sacred Community Places (SCP), we have opened the selection of places to community residents. In the same way that a Blacksburg user made a light pole significant, SCP users anoint places by being at those places. Any place, such as a picnic table or a bench in front of the library, can be significant. The user explains its meaning.

For example, one State College resident recalled “buying a turquoise teapot at an Arts [Festival] booth at the intersection of Allen and Fairmount Streets the first year we lived in State College - 14 years ago.” This recollection is accessible through a marker on a digital map. Other residents can access and interact with it by liking it, adding comments, and so forth.

Various personally significant places could be aggregated and displayed together in a heat map visualization, which could be seen as an embodiment of the town’s sacred structure. Many communities allow people to see and absorb heritage. However, fewer communities have programs for citizens to play an authorship role in heritage. In our work, we assume basic levels of data and technological literacy among participants. However, learning and development are important aspects of any community data initiative aimed at lowering citizen participation barriers. In the following section, we discuss ways in which communities undertake educational initiatives – both formally and informally – to strengthen citizens’ abilities to make meaningful civic contributions as well as efforts to strengthen knowledge and skills pertinent to community data.

3 COMMUNITY LEARNING AND DEVELOPMENT

A central role of community is informal learning. As described by writers like John Dewey [10] and Margaret Mead [25], children naturally participate and model practices in adult activity. For example, children watch parents build fires, convey caring through social interaction, or express concern about weather patterns. Children internalize these constitutive values and skills as part of an expanding sense of the interests, responsibilities, and capacities of being a member of a community.

This modeling shows how vital domestic practices as well as social and symbolic aspects of culture are transmitted from one generation to the next, and it is a way for children to distinguish between public and private roles. One way to explore the importance of this distinction is to consider its practical implications. If children cannot distinguish between private and public roles, then as they grow into adulthood they may act in ways that are privately beneficial and potentially detrimental to communities. A benefit of community apps like Lost State College and Blacksburg Nostalgia is that they reinforce the dual identity of their users.

On the one hand, users share personally meaningful stories about a community place of their choice or about a pre-determined significant place. Sharing a personal experience or perspective could be seen to communicate the value of an individual as *a member of a community*. Stories may not always match up because memories are personal and faulty. Other members of the same community may not agree on the significance or meaning of a particular communal place. But seeing – or better yet participating in – these moments of communal discord seem to us to be crucial events in the formative years of a community member.

A possible value proposition of apps like Lost State College or Blacksburg Nostalgia is that they provide a platform for modeling communal practices, including disagreement and debate over community values and heritage. To participate meaningfully in such modeling, however, requires knowledge of the community and relevant data as well as necessary analytic and technological skills.

While it is possible to observe and internalize familial values from a parent in the home, there are fewer opportunities to observe and internalize community values from neighbors and community members interacting around community heritage and local government policy and decision-making. Children and adults alike could benefit from increased exposure to this kind of community interaction.

Contemporary communities present different constellations of opportunities and requirements for citizen engagement that continue to increase in complexity. For example, our local community is engaged in the Heritage State College Project, which involves planning “how two of its most heritage-rich neighborhoods... could be preserved for future generations” [16].

This project is not the exclusive domain of local government actors. Active citizen involvement fuels its progress, and this involvement requires citizens to interpret and make sense of data related to historic preservation and to collaborate with an urban planning firm to design and iterate on different design proposals for the community. This entails learning about preservation strategies,

implications of preservation for residents, government processes, design processes, and so forth.

The Borough of State College, Pennsylvania, has taken time to educate citizens and, thus, empower them as participants in the decision-making process. For instance, as of the writing of this paper, they have held three workshops during which citizens have collaborated with design professionals and public officials. The outcome of these workshops is a proposal for a way forward, which will be presented to the Borough council to guide their decision to pursue (or not) preservation status.

Many opportunities for citizen involvement in local government decision-making, such as Heritage State College, center on learning and technology. Moreover, there is a lot of content to learn, and one challenge is to provide sufficient instruction so that community members can make meaningful contributions. For example, residents participating in Heritage State College must have some knowledge of historic building preservation practices in order to make informed contributions.

Similarly, community members engaged in water quality monitoring must know about water quality, water quality data, and proper methods and tools for data collection and analysis. This sort of learning can happen through formal educational initiatives, such as government-initiated workshops. However, informal educational channels might be just as effective. For example, the Pennsylvania Senior Environmental Corps (PeSAC) is a local chapter of a “countrywide organization of seniors who promote stewardship of the environment within their communities by monitoring and improving stream [water] quality” [29]. PeSAC members acquire pertinent knowledge and skills *informally* from each other thus strengthening their capacity for self-governance.

In 1996, our research group collaborated with a similar group, the Save Our Streams (SOS) initiative, in Blacksburg, Virginia, USA, to investigate how then-novel Internet-based collaborative tools could support the collection, analysis, and discussion of water quality data.

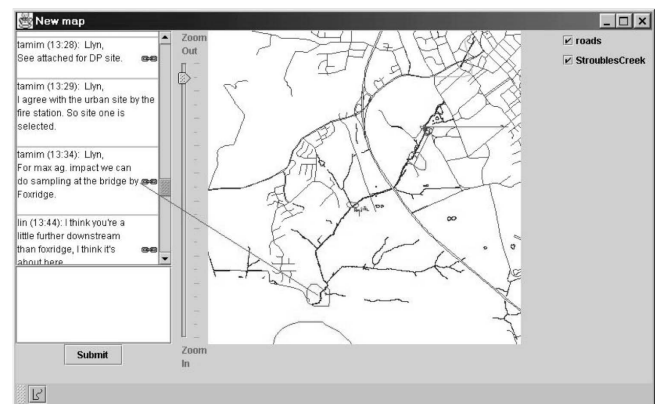


Figure 3: Save Our Streams interactive water quality data sharing tool.

Stroubles Creek, which originally ran through the center of Blacksburg, had been buried many years before 1996. Consequently, it

had become a distressed stream through its reduced access to air and sunlight.

We created the tool (Figure 3), which visualized the stream's route on a map and provided a way for the members of SOS to post and discuss water quality data; anchoring them to specific locations along the route of the stream. The visualization of the stream was itself revelatory since the stream is buried, and, thus, invisible in many places. Moreover, it provided fodder for asynchronous discussions about data collection and preservation strategies, which took place within the tool.

Today, this system suggests many possible elaborations. For example, it would be possible to denote different contaminant concentrations with color coding. Higher intensity regions could be distinguished with different color saturation levels. Colors would be linked to specific contaminants in order to convey spatial distribution and possible interaction. In addition, water recharge areas and places where water is drawn for use could be called out to emphasize both the threats to water users and the opportunities to strengthen natural methods of amelioration.

Members of SOS were well versed in water quality data, and yet water quality is a relevant issue for community residents who lacked the kind of knowledge or skill required to participate in a community data project geared towards water quality monitoring and assessment. To broaden the scope of community involvement, to achieve more pervasive participation, which is of interest to us both as researchers and as members of a community for which water quality is a pertinent issue, would thus require significant formal and informal learning and development initiatives.

Consider the mobile app, Lost State College. The community data infrastructure to probe and identify sacred structures requires significant new learning about the operation of mobile internet devices, management of digital data, and the use of new codes such as comments, replies, likes, and so forth. Some of this learning can come from community institutions.

For example, libraries might sponsor workshops and panel discussions enabling community members to acquire the technical skills required to use such an app. Moreover, local governments may facilitate expert panel discussions to teach citizens about environmental issues. And, in some cases, community members themselves may volunteer their time to teach others. But information infrastructures have transformed generation to generation through the 20th century. Now, they evolve even more rapidly. How will people learn and evolve with them? While formal educational initiatives, such as local government-sponsored workshops or library classes, could play a valuable role in learning and development processes, we argue for more informal learning and development opportunities through initiatives like PeSAC, SOS, and others. It might even be feasible for residents to adopt neighborhood data ambassador roles in an effort to facilitate more opportunities for informal learning about community data.

Community data interactions require data sophistication on the part of community members. It is important for members to understand different kinds of data and possess the knowledge and skill to work with and analyze it in different ways. But complex data interactions can also provide concrete and motivating contexts for developing skills and concepts pertaining to data.

Up to this point, we have conceived of community data as a way to reinvigorate and strengthen a set of community goals, including: developing personal and collective identity, enabling awareness and participation of community members, and facilitating social learning and development. In addition, we have emphasized possibilities for enhancing data literacy within local communities. Taken together, these possibilities inspire us to conceive of a new paradigm for local deliberation, decision-making, and self-governance; one that is grounded on shared data.

4 COMMUNITY DELIBERATION, DECISION-MAKING, AND GOVERNANCE

As our Neolithic ancestors cooperated to plan, build and operate shared villages, we can cooperate to identify, analyze and interpret our own shared data, reinvigorating and strengthening democratic institutions and practices from the bottom up. In this section, we explore community data as a means of strengthening community members' participation in civic deliberation and decision-making processes.

Arnstein developed the concept of "citizen participation" [1], and, since its inception, several extensions and variations of this concept have been explored. The field of public administration is rich with theoretical and empirical examples of citizen involvement in communities. Vigoda [37] has put forth the need to move from citizen participation to collaboration and heralds mutual effort between citizens and governments as the next generation of public administration. The digital age, as Linders [21] describes, has provided opportunities for citizens to partner with the government instead of being subjects of the government. Given the role of citizens and their associated community data then becomes central to community deliberation, decision making and governance. Some scholars have envisioned public service as being co-produced, wherein service providers and users make balanced contributions at each stage from planning to evaluation [2, 3]. Community data can be a vehicle for this co-production process since enables citizens to contribute meaningfully in local deliberation and decision-making processes.

For example, household energy data collection and analysis tools have made it possible for community members to understand home energy use patterns. Froehlich et al., [12] found that when people understand these patterns, it becomes possible to change home energy use behaviors, which, in some cases means reducing their overall energy footprint. Preliminary research also suggests that comparing individual energy use patterns with community-level household data yields a deeper understanding of personal use than would be achievable *without* community data.

Encouraging community-level participation in collecting and sharing household energy use data may offer value beyond the individual level. Access to community energy use data puts local residents in positions to judge when to use heavy-consumption appliances. For example, they can run a dishwasher during off-peak times so to avoid taxing the power grid. In addition, energy providers can identify energy-related issues and better distribute energy based on the community usage patterns [26]. For example, providers can distribute electricity based on peak and off-peak times to avoid power outages. Finally, local governments can learn from

local household energy data to make decisions about sustainable practices at the community level. Household energy usage data is an important kind of community data, but modern communities and individuals may have a more active role in collecting and analyzing other kinds, such as: demographics, economic data, and air and water quality data.

Different kinds of data afford different opportunities for coproduction at the community-level. For example, although household energy use data is useful, community members cannot manage their local energy sourcing. In contrast, where water quality data is concerned, local residents may have the capacity to verify and exert control over what comes out of the ground.

For example, community residents in State College recently made use of water quality data, land-zoning proposals, and geological surveys to advocate for a water and farmland conservation campaign. They found that a proposed land development project would lead to storm water accumulating harmful toxic materials en-route to the community drinking water supply due to the site's proximity to an intermittent stream and two important wells. Residents used community data (including historical events related to the land and rezoning documents) to argue for land preservation. Additionally, they generated a knowledge base of relevant data in collaboration with local domain experts.

This knowledge base reaffirmed the importance of pervasive participation in community data initiatives. More community members collecting more water quality data over longer periods of time enables citizens to recognize factors that could put water quality at risk (e.g., the proximity of a land development site to a community drinking water source) as well as subtle changes in the data – i.e., changes in relation to an established baseline – that may be cause for alarm. Community data needs to be legitimated, curated, and preserved. It has to become part of the community's heritage, and it must be remembered.

As far as we know, none of the community members involved were expert hydrogeologists, but they gained sufficient working knowledge of relevant issues, including: water quality, storm water management, farmland irrigation, and so forth. In our conversations with them, we learned that they had identified several issues with the land developer's data. We see this as a crucial affirmation of the value of data literacy. Without data literacy it would be difficult (if not impossible) for citizen participants to assess the strengths and weaknesses of a particular data set. Data literacy empowered some community members in this case to make a stronger argument for preservation rather than development.

Community members had a voice in this decision-making process, which is valuable in and of itself. In addition, their voice may have been stronger and more effective because of their data literacy. This highlights a crucial link between community data and public deliberation and decision-making. While the residents in this case largely took the initiative to collect and analyze data – or seek expertise for data that was too esoteric – other cases highlight the role community data plays in collaborations between local governments and citizens.

PlaceSpeak (Figure 4) is a web-based platform to encourage civic engagement in local communities [30]. Local governments post proposals or seed discussions about relevant topics, such as: urban

planning and local heritage preservation. Verified current local residents can participate in discussions. PlaceSpeak thus provides a digital community space in which citizens and local government officials engage in civic discussions. Contrary to other social platforms on which local governments interact with their citizens, PlaceSpeak disallows participation from non-members of the community. This focused view of who counts as a member of the community can be beneficial.



Figure 4: Visualization of the PlaceSpeak transportation vote in Vancouver, BC, Canada.

Exclusive discussions empower current residents to learn about and contribute to the decisions that impact their lives and local communities [34]. For example, the Vancouver government collected votes for a local transportation and transit plan and people's satisfactions about the voting outcome via PlaceSpeak [38]. The outcomes clarified how community members felt about the new plan as well as how they felt about the voting outcome. The Vancouver government was able to implement the transportation plan based on the votes and people's satisfactions.

By participating in these kinds of discussions, residents can gain access to new data and information about possible changes to their community, which could contribute to a stronger sense of local government transparency and an increase in civic participation. Moreover, they can learn about what kind of data their local government collects or has access to and how it analyzes and interprets that data in the service of policy decisions. In addition, through these discussions, local governments can learn about their citizens existing knowledge of and ability to work with data, and about what additional support could be provided to enhance the quality and quantity of citizen participation in deliberation and decision-making.

Here, trust and data literacy become key issues. Community-government communication and co-operation hinges on the concept of trust. Citizens must trust a local government's presentation and provision of data. *Is it accurate? Biased? Has it already been manipulated in some way?* Moreover, they must possess the knowledge and skill to assess the trustworthiness of a particular data set on its own terms. They must know the right questions to ask of data so that they can establish trust in its utility and value.

One way of establishing data trustworthiness is in making available any data *about* data (i.e., metadata), which has been referred to as "provenance" [40]. Secure provenance and auditing of provenance in networked and secure environments can establish trust

between citizens and data, and we see this kind of trust as an enabler of cooperation between citizens and government. Community data is in a unique position to establish trust and enable citizens to partake in meaningful deliberation and ownership of policy decisions. Community and local government ties are inherently related to public policies and, in turn, to evidence in the form of data that they bring to bear. Community data offers a reciprocal pathway between community members and the local government. It presents an opportunity for local government bodies to showcase their commitment to transparency as well as their capacity for equitable policy and decision-making. Moreover, it is a mechanism to increase the trust base in the local government.

Keele [20] argues that engagement in civic life is an important factor in macro-level trust in government. Technological developments that may undermine efforts to establish trust ought to be examined and critiqued. For example, a paradoxical trend has characterized the discipline of data analytics in recent years. Autonomous/smart predictive algorithms are becoming more incomprehensible. This trend is referred to as the “black-box” of analytics. We can imagine scenarios in which these algorithms might perform the kinds of community data analysis that we have been discussing in this paper and, thus, negating the need for local governments and their citizens to devote time and resources to data analysis in the service of community deliberation and policy-making. But this black-boxing of civic decision-making processes ought to be avoided, and we believe that it is possible to avoid it with the help of community data.

Citizen participation via community data has potential to strengthen local democratic processes. Opening the “black box of data analysis” can have transformative capabilities; advancing citizens from being mere data points or sensors to data analysts, and ultimately moving towards an ideal state of self-governance.

5 DISCUSSION

Local community offers diversity and scale for investigating and empowering pervasive participation in human society. Indeed, community as “living laboratory” was the original research model for American sociology [28]. Our work builds upon the sociological concept of the living laboratory regarding place-based structures and interactions as central to the natural ecology of social life. However, we also emphasize participatory interventions, specifically community data activities, such as co-curating collections of sacred community structures, telling personal stories about shared community spaces and events, tracking water quality data as a way of participating in local government decision-making, and so forth.

The community is a living laboratory comprised of local actors, institutions and activities that create a robust, coherent and dynamic context for exploring and assessing new ideas through long-term participatory learning [4, 7]. The centrality of individual and local group agency and responsibility, and the scope and degree of routine innovation in our conception of community as living laboratory, contrasts with Park’s earlier sociological notion.

Tönnies [36] argued that the Industrial Revolution undermined the notion of community, thus initiating a discourse of “lost community,” which asserts that various technologies undermine personal

intimacy, trust, empathy, collective identity, participation, and support networks [31, 33, 35]). Such critical perspectives are useful in helping us pose analytical questions. For example, Tönnies’ concept of community was feudal and suffocating. The new cities of the industrial age replicated most of those inequities and added new ones. His critique is nuanced from a modern perspective.

Sennett [33] takes a different approach. He argues that “public life,” which in our view is analogous to community life, has degraded in part because crucial mores that separated private from public life – ones that supported and promoted activities such as civic debate over important community issues – have been stripped away.

In Sennett’s view, people made efforts to cultivate public relationships as warm and intimate in a manner one might expect from family or close friends. “The expectation is that when relations are close, they are warm; it is an intense kind of sociability which people seek out in attempting to remove the barriers to intimate contact, but this expectation is defeated by the act” [33, p. 338]. What happens when public interactions do not provide the kind of warmth and intimacy that people may have come to expect? In many cases, community members withdraw into spheres made up of family and close friends. Putnam [32] calls this “hibernation.”

Some forms of community engagement require a level of dispassionate rationalism. Attending town council meetings and participatory design workshops require community members to recognize their position as parts of a whole community. This means engaging with, listening to, and at times subjugating one’s own views and needs for a greater purpose. In a democratic community, it means coming up against people with different, conflicting, and at times offensive or hurtful views and opinions.

Our vision of pervasive participation in community data collection and participation is one of a culture of critical debate, a more fearless trust between community members, and trust in facts on the ground (and *in* the ground). These qualities are crucial for communities to survive and thrive, which means they could be considered part of a possible definition of collective community efficacy. Consider a statement like, “I believe my community can manage local resources by identifying relevant data and critically debating its interpretation.” Every single person in a community does not have to participate in identifying data and debating its meaning. But the community’s identity and commitment to participation are sufficient to value such activities and ensure that they happen.

Design and technology provide a complementary perspective, envisioning new possibilities for human cooperation, new opportunities to rekindle a strong sense of community, and sustainable ways of living. Emerging sociotechnical forces can be transformed into resources for strengthening trust, collective identity, participation and support networks.

For example, we can engage community members in learning and development by collaborating with local institutions, like public libraries, as satellite data repositories and sites for learning about data collection and analysis. Libraries provide computers and Internet access to patrons, and, crucially they also provide support staff.

Library staff could function as community data ambassadors; assisting patrons with basic technology as well as teaching them about community data. A supportive, proficient data ambassador

might be especially valuable to members of the community who lack proficiency working with numbers. In informal discussions about community data with local research scientists, we have found innumeracy – a lack of literacy with numbers – to be a commonly cited roadblock to working with data.

While citizens may feel compelled by a topic of interest and importance to a community, such as drinking water quality, confronting a numeric data set may overpower their will to act. A lack of data literacy could thus be seen to motivate people to work *against* their own interests.

To better integrate and legitimate community data, local governments could organize formal, quarterly hackathons where people with diverse sorts of expertise gather together and work on real problems through community data analysis and data-driven argumentation.

In our own work, we have thus far facilitated two community hackathons involving local water quality stakeholders in the service of designing an integrated water quality data platform. We discussed explored possible sources and types of water quality data that might be useful and about different stakeholders who might use such a platform, including: sharing stories about local water resources, school children analyzing and sharing water quality data sets, and so forth. We hypothesize that involving local stakeholders in the design of an open data platform could be an effective way to encourage interaction with the data that (will be) open to them. In addition, dedicated systems for storing and sharing community data would render themselves useful in creating a stable, reliable destination for citizens to access and work with community data. Identifying other arenas of community data and investigating how those data are gathered and used could be part of such a research agenda going forward.

Identifying and capitalizing on opportunities for informal learning are also ways to address this problem. Establishing data ambassadors at libraries and in different residential neighborhoods could be a useful approach. Ideally, communities will reach a point where residents do not have to seek out formal channels for learning about data collection and analysis. They will either possess the necessary knowledge or skills to work with community data or they will know which members of their community perform informal data outreach.

Community data increases social awareness of heritage and values while also making shared experiences more engaging, personal, and real for individuals who share the experiences and data. As such, we can more clearly see that community exists through time and not just across a space. Heritage and values are preserved through community data and they can facilitate our comprehension as to why certain issues exist or are engaging as well as how a community obtains a collective community identity.

Community data also offers ways for understanding community problems, weaknesses, strengths, and opportunities. This is significant because community data can be used to strengthen bonds and trust between individuals and groups within the community through informal and formal learning opportunities. These bonds can be leveraged collectively to work towards a community goal and to democratically create governing policies that are suitable and acceptable to diverse members of a community. It also empowers local community members to rally support for (or question) the

current position on a particular community issue. Community data thus has tangible effects upon individuals and communities.

Community data can be strengthened through greater data literacy among diverse community groups and through greater consolidation of resources within and between those diverse groups. Community data is most meaningful to a community when it accurately reflects community values and heritage. As such, promoting data literacy amongst all individuals of the community should be encouraged, although such actions may be unnecessary for representing community opinions.

Currently, community entities offer informal and formal learning opportunities to individuals, and individuals often already have social ties. Training a group of individuals and utilizing their existing social ties could allow them to represent groups within the community more adequately. Pervasive participation in community data is important, but that does not mean everyone participates all of the time. The key is the collective.

6 CONCLUSION

In this paper, we argued that community data (i.e., hyperlocal data that describes the community) is gathered, analyzed, and used by the community. It provides opportunities for and benefits from pervasive participation. However, availability and access to community data are not enough to achieve pervasive participation. Other complementary initiatives, such as formal and informal learning, community data workshops, and other activities are necessary. Because data and data handling are a frontier of information technology, community data can be framed as a frontier of community innovation and of the continuing reinvention of community in modern society.

Referring to design and development projects from the past two decades, we showed how data can codify and focus community values and heritage in data-driven activities of identifying, analyzing and discussing the sacred structure of the community as personal and diverse place-based meanings. Projects like Blacksburg Nostalgia, Lost State College, and Sacred Community Places envision a pervasive participatory re-imagining of what the community is, integrating the past, present and future of community places. More recently, we discovered a viable, healthy, and citizen-driven data collection practice in our local community that we wish to support and strengthen through the development of an integrated water quality data platform. Crucially, our community is not an isolated example of such practices. However, it is one “type” of community, and there are others where a data-driven approach could apply.

There are communities across a spectrum from impoverished to metropolitan whose members do not worry about water quality; they worry about water *access*. Cape Town, ZA, is a contemporary, well-known example. But there are many places suffering from water shortages and drought, including: Jakarta, Melbourne, Mexico City, São Paulo, various places in India, as well as the Southwestern United States. In all of these cases, water management policies *not* based in data are at the root of the crises [39].

We described how community data can motivate and guide community-wide learning about the appropriateness and utility of different kinds of data and data analytic approaches. The literacy skills of citizens have always been a key enabler of meaningful

participation in community and society. Literacy has traditionally been equated with reading and informal logic. As we move into an era mediated by data and data-driven thinking, literacy can expand to include the design of data-oriented investigations, the interpretation of data, and the development of municipal visions and policies grounded in community data.

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